

Research Statement

Gustavo Perez

I am a computer vision and machine learning researcher. My research aims to make foundational contributions towards developing methods for extracting measurements from large datasets to enable fundamental discoveries in interdisciplinary science. For instance, how climate change influences bird phenology [3] or how a galaxy’s star formation rate is related to its intrinsic properties [5].

Despite the potential of AI to accelerate scientific discovery, current computer vision systems face significant limitations, including poor performance with limited training data, potential biases, and the lack of statistical guarantees, often requiring manual verification of model predictions by specialists.

With advances in computer vision, machine learning, and statistical estimation, as well as the incorporation of human feedback, my goal is to enable *accurate scientific measurement in large datasets*. My research can be divided into two complementary threads: First, **in the context of model development**, I investigate methods for robust and data-efficient learning by integrating classical filtering principles into modern deep learning [10], I propose techniques to improve knowledge transfer to novel domains [9], and develop tools with unsupervised learning and human-in-the-loop for data-limited settings [12]. Second, **in the context of model deployment**, I combine AI and statistical estimation to reduce the manual verification effort of model predictions. I derive estimators for various scientific metrics, including the number of objects in large image collections [13] and the population size from visual re-identification tasks [14].

Most of my core AI research is motivated by relevant scientific problems involving interdisciplinary collaborations with findings published in leading journals across ecology, astronomy, and chemistry. For instance, I work with ecologists to study phenology in bird species using radar imagery [1, 3, 4, 15, 17], with astronomers to provide insights into star formation using data from the *Hubble Space Telescope* (HST) and the *James Webb Space Telescope* (JWST) [5, 6, 11, 12], and with chemists to predict absorption in nanoporous materials for energy-efficient chemical separations [7, 8].

Model development for scientific discovery with AI

Improving deep learning with classical wisdom [10]. Classical image filters are carefully normalized to ensure consistency, interpretability, and to avoid artifacts like intensity shifts, halos, or ringing. In contrast, convolutional filters learned end-to-end in deep networks lack such constraints. I address this limitation by proposing filter normalization, a simple yet effective modification that ensures the filters are atmosphere-equivariant, enabling co-domain symmetry. Our analysis reveals that unnormalized filters degrade performance, whereas filter normalization regularizes learning, promotes diversity, and improves robustness and generalization (Fig. 1).

Improving knowledge transfer to novel domains [9]. Transferring deep networks trained on large datasets of color images has been a key to their success in visual recognition. However, a challenge arises when transferring to heterogeneous domains where some architectural modification is necessary to process the input.

I consider the problem of adapting a network trained on color images to hyperspectral images with a large number of channels and propose a novel multi-view scheme that generates a prediction by aggregating information across different *views* of the input (Fig. 2). Our proposed multi-view adaptors outperform simple and effective schemes like linear mappings with negligible added parameters.

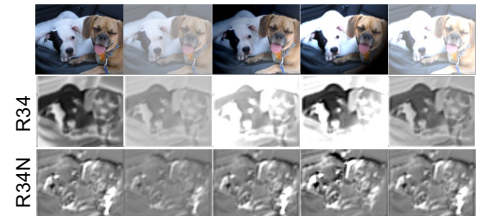


Fig 1. The response of a ResNet34 with filter normalization (R34N) remains more robust and informative than without normalization (R34) to intensity transformations (top row).

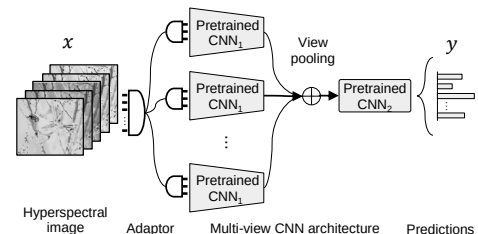


Fig 2. Multi-view domain adaptor.

Representation learning with limited data and human-in-the-loop [12]. In this work, we focus on AI tools for cataloging bright sources, such as star clusters, within galaxies. This process requires significant domain expertise and resources. To accelerate this step we propose a web-based annotation tool to label and visualize high-resolution astronomy data, encouraging efficient labeling and consensus building, and techniques to reduce the annotation cost by leveraging recent advances in unsupervised representation learning on images (Fig. 3). We present case studies where we work with astronomy researchers from UMass Amherst and Stockholm University to validate the annotation tool and find that the proposed tools can reduce the annotation effort while facilitating accelerated analysis of new data.

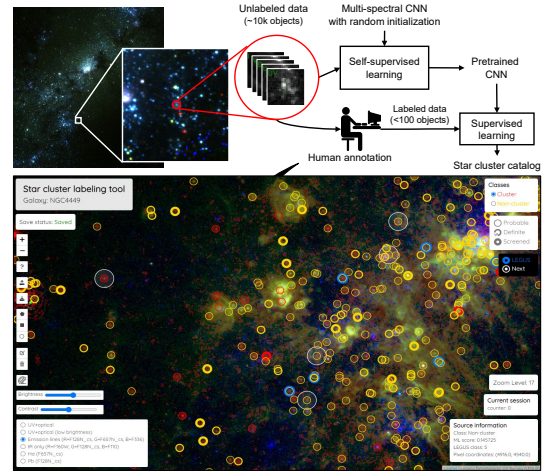


Fig 3. Proposed AI-assisted tool.

Future research. Specialized architectures tailored to scientific domains are crucial for achieving meaningful and efficient performance, as generic models designed for RGB images often fail to capture the physical and statistical structure of domain-specific data. Moving forward, my goal is to develop hybrid models that bridge data-centric and knowledge-centric AI, combining the flexibility of modern deep learning with the interpretability and physical grounding of scientific theory. By explicitly embedding domain knowledge—such as invariances, symmetries, or physical constraints—into model design, we can create architectures that are inherently consistent with the underlying science, reducing data requirements and improving generalization. I am also interested in exploring how such knowledge-guided designs could interface with large vision-language models, which may offer new avenues for cross-domain transfer and rapid data interpretation. Could these models, when aligned with scientific priors, enable more efficient hypothesis generation or surpass ImageNet-based pretraining in heterogeneous, data-sparse domains?

AI and statistical estimation for scientific measurement in deployment

Counting in large image collections [13]. Many modern applications use computer vision to detect and count objects in massive image collections, which are too large for humans to process in the available time. So, a common approach is to train a detection model and run it exhaustively on the images. A particular challenge occurs when the detection task is very difficult, so the accuracy of counts made on the entire collection is questionable even with huge investments in training data and model development. Some works resort to human screening of the detector outputs, which saves time compared to manual counting but is still very labor-intensive.

In DISCOUNT, we propose a detector-based importance sampling framework for counting in large image collections that integrates an imperfect detector with human-in-the-loop screening to produce unbiased estimates of counts. We propose techniques for solving counting problems over multiple spatial or temporal regions using a small number of screened samples and estimating confidence intervals (Fig. 4). We demonstrate our method with two applications: counting birds in radar images

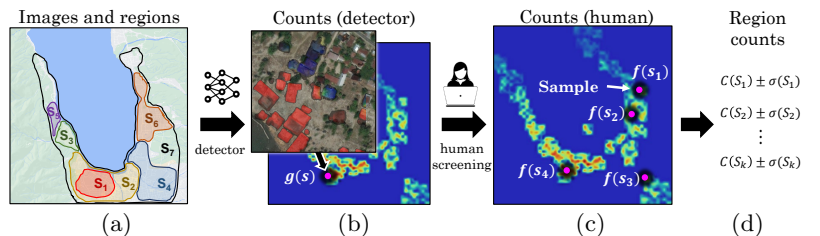


Fig 4. DISCOUNT uses detector-based importance sampling to screen counts and solve multiple counting problems. (a) Geographical regions where we want to estimate counts of damaged buildings. (b) Outputs of a damaged building detector on satellite imagery, which can be used to estimate counts $g(s)$ for each tile (shown as dots). (c) Tiles selected for human screening to obtain true counts $f(s)$, from which counts for all regions are jointly estimated by DISCOUNT.

to study responses to climate change, and counting damaged buildings in satellite imagery for damage assessment in regions struck by natural disasters. The work was awarded the Best Paper at AAAI-AISI'24.

Estimating population size in visual re-ID tasks [14]. We extend DISCOUNT to estimate cluster counts via pairwise similarity, enabling population size estimation with imperfect re-ID systems and minimal human vetting. Our nested importance sampling approach outperforms existing active clustering methods on various animal re-ID tasks.

Future research. My research focuses on integrating human feedback into AI models for scientific measurement. I will explore various approaches, including active learning, semi-supervised learning, or direct statistical estimation, and continue to collaborate with domain experts to better understand the trade-offs and develop effective solutions for different domains. Beyond counting tasks, my work shows that combining AI and statistical estimation can derive various scientific measurements, such as those in [14]. This approach could be extended to other applications, like estimating astronomical correlations or interactively improving models with the acquired labels during the screening process.

Applications

Ecology. Weather radars are sensitive enough to detect flying animals, providing opportunities to study roosting behaviors at large spatial and temporal scales. However, extracting biological insights from the data is challenging, limiting the scope of ecological research.

In collaboration with aeroecologists from Colorado State University and The University of Oklahoma, our work investigates the impact of environmental change on aerial insectivores' seasonal timing and population. We propose a computer vision system (Fig. 5) to detect roost signatures in radar data and a pipeline for large-scale processing [15, 17], providing research-quality data with far less human effort than manual annotation of radar scans. Our successful deployment on 8% of the radar stations in the contiguous US lays the groundwork for continent-scale analysis of swallow roosts and provides a starting point for analysis of other family-specific phenomena in weather radar, such as bat roosts and mayfly hatches. This data has led to several publications [1, 3, 4, 2].

Astronomy. Star cluster properties, such as age and mass, provide insights into the process of star formation. However, significant domain expertise and resources are required to discriminate star clusters among tens of thousands of sources in a galaxy. In collaboration with astronomers from UMass Amherst and Stockholm University, we propose STARNET [11], a classifier of star clusters in high-resolution images of nearby galaxies captured by the HST. We later adopted STARNET to analyze star cluster formation in galaxy M101 [6], where it reproduced human classifications with high levels of accuracy.

Recent data from the JWST is set to transform our knowledge of star formation by observing young star clusters previously obscured by dust and gas. Our work in [12] showcased the tool's potential in exploring these obscured sources to study relationships between star formation rate and infrared emissions [5].

Chemistry. Energy-intensive chemical separation processes consume over 10% of the global energy. Nanoporous materials offer a promising alternative, but identifying the best material is a significant challenge. In collaboration with UMass Amherst chemical engineers, we develop AI techniques to identify optimal materials for specific separation tasks, with a focus on nanoporous zeolites [7, 8].

Medical imaging. Motion-related artifacts remains a major challenge in magnetic resonance imaging (MRI), degrading image quality and diagnostic accuracy. Traditional motion mitigation strategies are limited in their applicability, particularly for high-resolution 3D scans that require longer acquisition times. We present a framework for motion correction in brain MRI with a MIMO approach using Pilot-Tone-based RF signals. We collected a dataset of 13 volunteers and demonstrate that our approach can robustly translate RF feedback to motion parameters on various acquisition sequences.

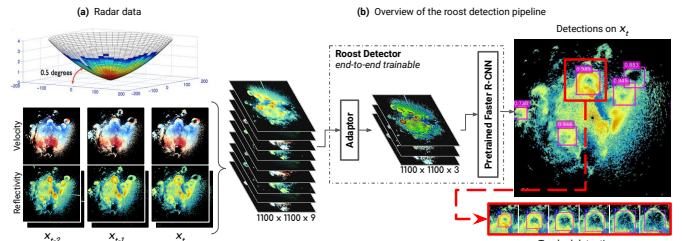


Fig 5. Using our model [15], we can identify birds, bats, and insects on weather radars, allowing for in-depth studies of specific species.

Civil engineering. Accurate building attribute extraction is crucial for effective urban planning, disaster response, and infrastructure management. However, current building attribute data collection relies heavily on in-person surveys and localized government records, which are expensive and often limited in scope and geographic coverage. To address these challenges, we propose a general workflow for scalable zero-shot attribute extraction that exploits recent advances in large vision and language models [16].

Future research. Interdisciplinary collaborations are a crucial aspect of my research. I plan to continue fostering interdisciplinary collaborations, advocating for the integration of AI across diverse scientific domains, and to better understand field-specific trade-offs to develop effective solutions.

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